Machine Learning Fundamentals

Week 1

Task 1 – Exploratory Data Analysis (EDA)

Dataset: Titanic - Machine Learning from Disaster

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Track: Neurofive Machine Learning Track

Objective

The objective of this notebook is to explore the Titanic dataset, understand its structure, identify missing values, distinguish numerical and categorical features, and summarize the dataset before applying any machine learning models.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Importing the Dataset

You can use this instruction df = pd.read_csv("train.csv"), only if Python can find train.csv in the current working directory (the folder from which Jupyter is running). Means Week1_Titanic_EDA.ipynb and train.csv are in the same folder.

But if the dataset and the notebook are in different folders then ue the following approach.

```
df = pd.read_csv(r"D:\Machine Learning Fundamentals Course\Week 1\
titanic\train.csv")

df.head()

   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age
SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0
```

```
1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0
1
2                               Heikkinen, Miss. Laina  female  26.0
0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0
1
4                             Allen, Mr. William Henry    male  35.0
0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

```
df.describe()
```

```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000
```

```
            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

```python
print(df.shape)
```

```
(891, 12)
```

```python
df.isnull().sum()
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```python
df.select_dtypes(include=np.number).columns
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Age', 'SibSp', 'Parch',
'Fare'], dtype='str')
```

```python
df.select_dtypes(include=['object', 'str']).columns
```

```
Index(['Name', 'Sex', 'Ticket', 'Cabin', 'Embarked'], dtype='str')
```

Data Story

The Titanic dataset contains information about 891 passengers and 12 different features. The dataset includes both numerical and categorical variables such as age, fare, gender, passenger class, and embarkation point. Some columns, including Age, Cabin, and Embarked, contain missing values that will require preprocessing before training machine learning models. Overall, the dataset appears well-structured and provides sufficient information for predicting passenger survival.

Task 2 - Data Cleaning & Visualization

After understanding the dataset through Exploratory Data Analysis (EDA), the next step is to prepare the data for Machine Learning. This includes handling missing values, identifying potential outliers, and visualizing the dataset to better understand relationships among features.

The objective of this task is to clean the Titanic dataset, create meaningful visualizations, and summarize the insights obtained from the analysis.

Handling Missing Values

Before building any Machine Learning model, missing values must be addressed because they can reduce model performance and may even prevent certain algorithms from working correctly.The assignment requires using fillna() or dropna() and justifying the choice.

First, we inspect the dataset to identify which columns contain missing values.

```
df.isnull().sum()
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
df["Age"] = df["Age"].fillna(df["Age"].median())

df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])

df["Cabin"] = df["Cabin"].fillna("Unknown")
```

Why was `fillna()` used?

Instead of removing rows using `dropna()`, missing values were filled because dropping rows would reduce the size of the dataset and could result in the loss of useful information.

The median was used for the **Age** column because it is less affected by extreme values than the mean. The mode was used for **Embarked** because it is a categorical feature. Missing values in **Cabin** were replaced with "Unknown" since most cabin entries are unavailable. Now Verify the Missing Values again.

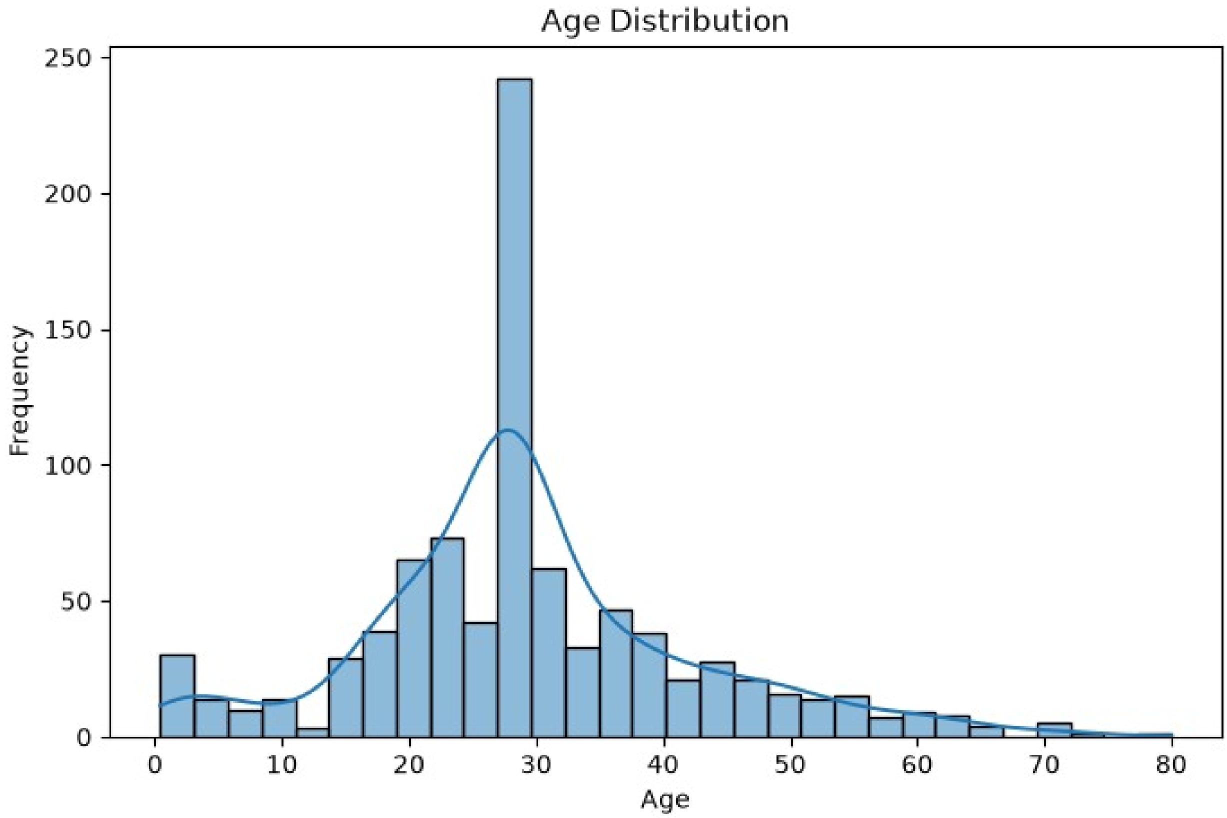
```
df.isnull().sum()
```

```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

Histogram

A histogram illustrates the distribution of passenger ages.

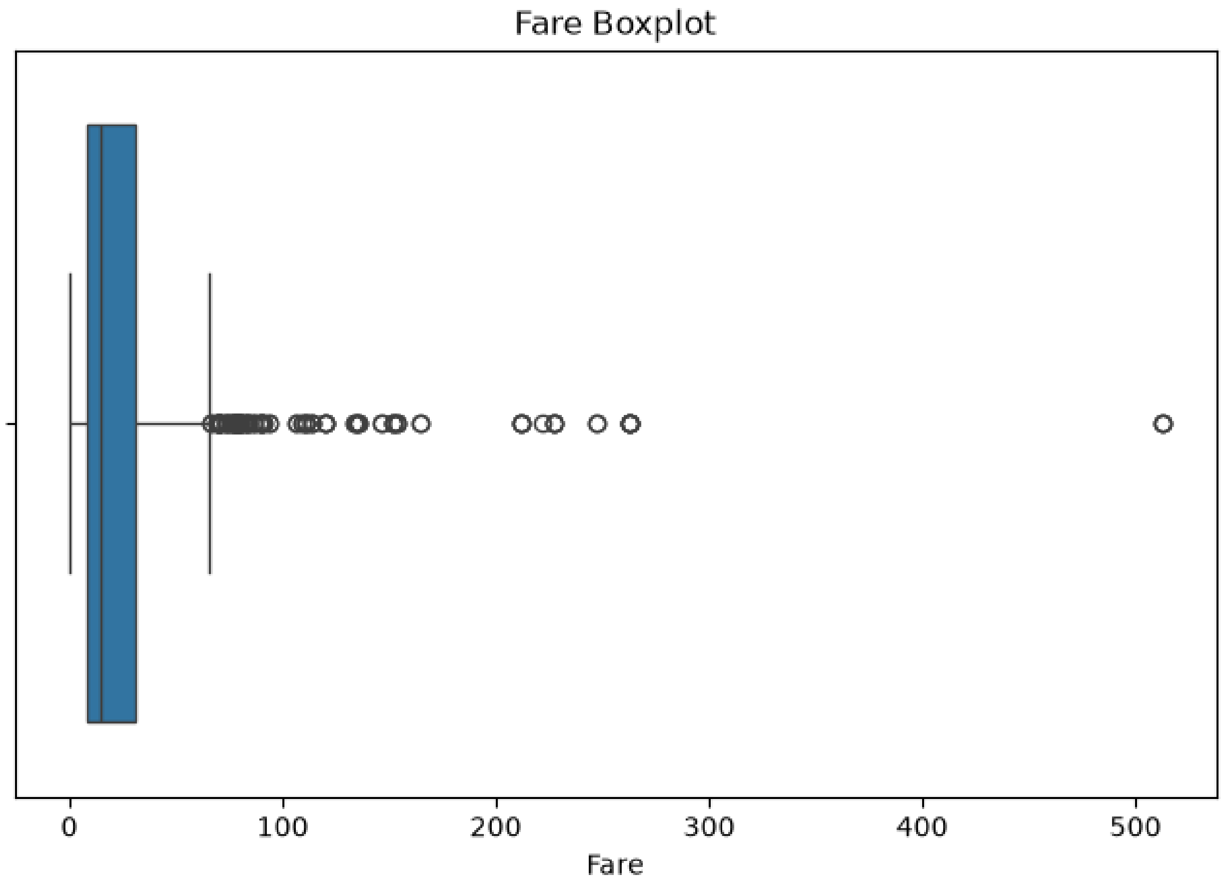
```
plt.figure(figsize=(8,5))
sns.histplot(df["Age"], bins=30, kde=True)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```



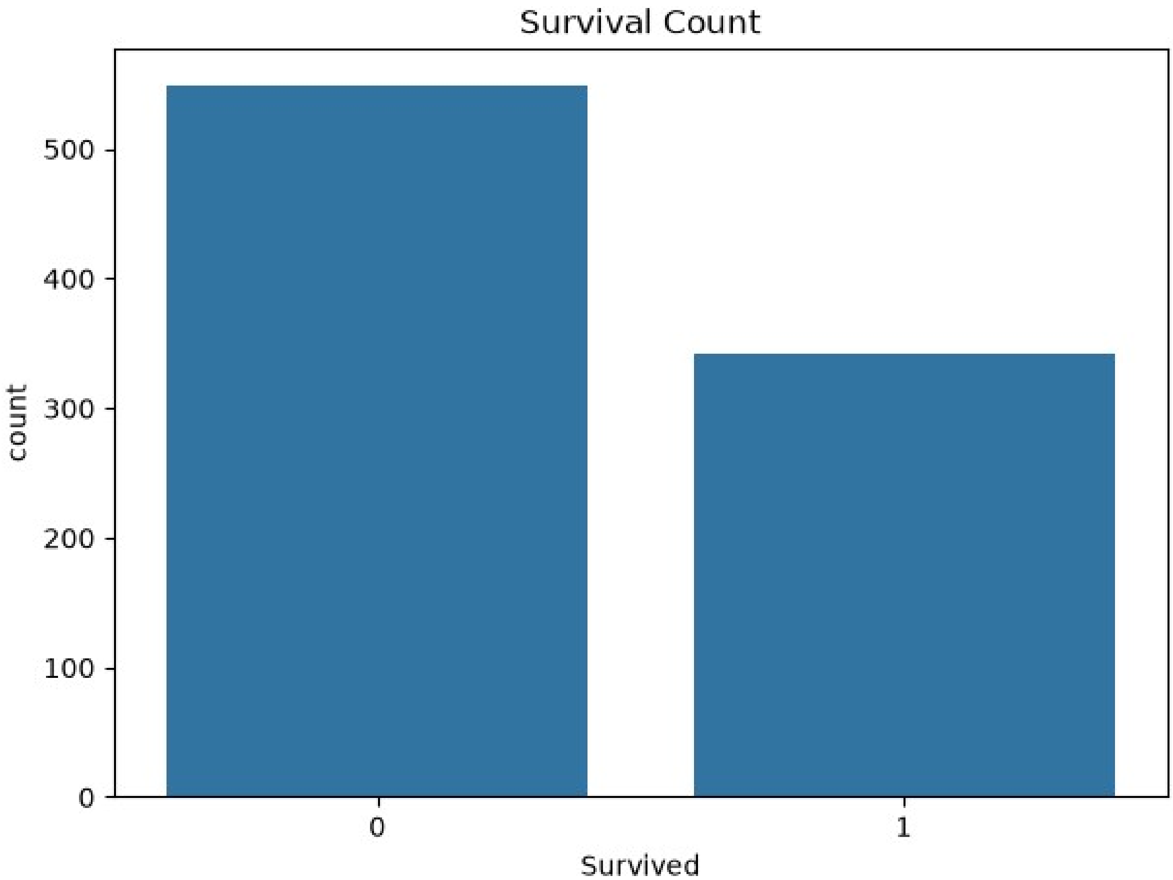
Outlier Detection

A boxplot is used to identify potential outliers in passenger fares.

```
plt.figure(figsize=(8,5))
sns.boxplot(x=df["Fare"])
plt.title("Fare Boxplot")
plt.show()
```

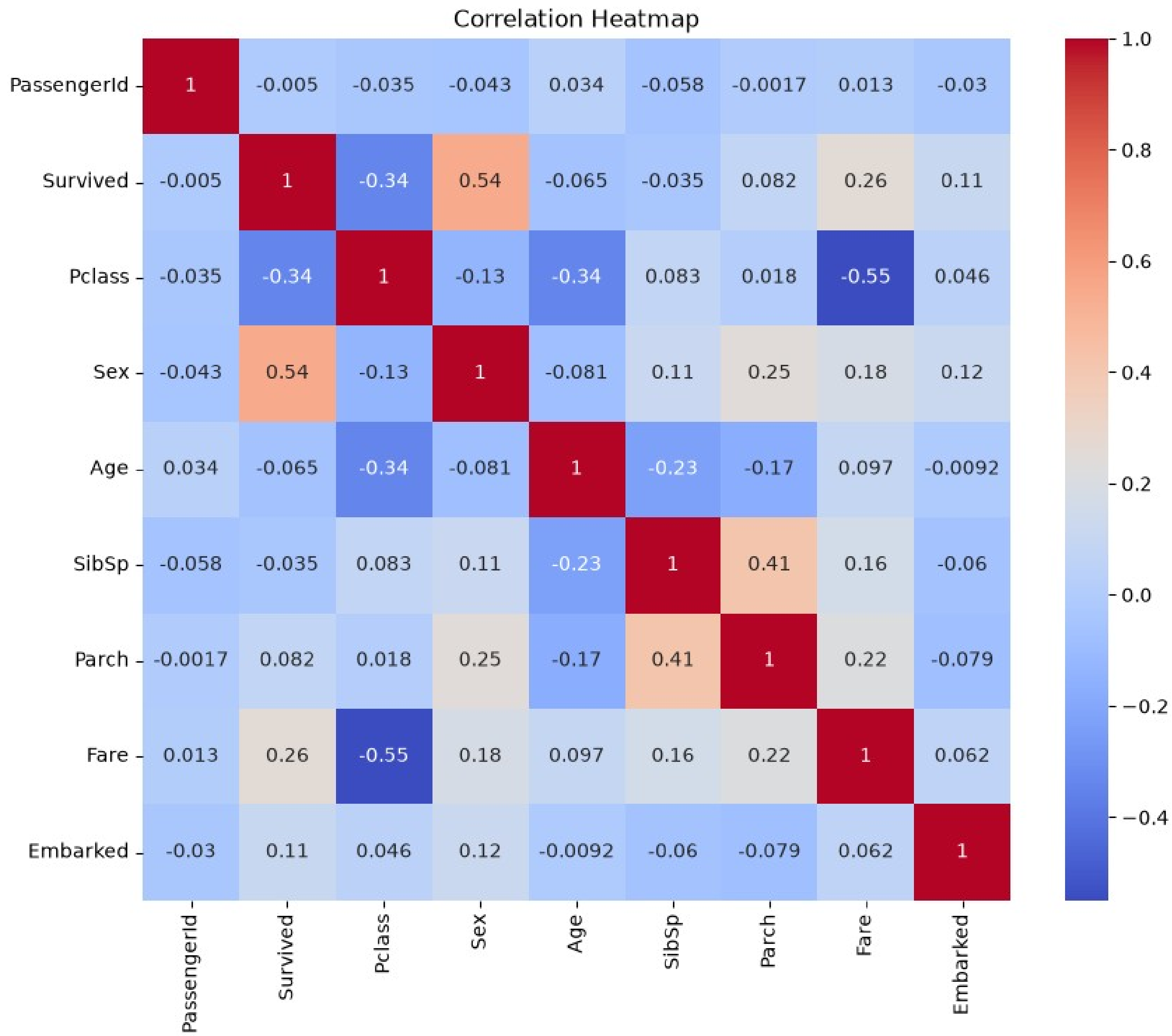



```
plt.figure(figsize=(7,5))
sns.countplot(x="Survived", data=df)
plt.title("Survival Count")
plt.show()
```




```
df_encoded = df.copy()
df_encoded["Sex"] = df_encoded["Sex"].map({"male":0,"female":1})
df_encoded["Embarked"] =
df_encoded["Embarked"].map({"S":0,"C":1,"Q":2})

plt.figure(figsize=(10,8))
sns.heatmap(df_encoded.corr(numeric_only=True), annot=True,
cmap="coolwarm")
plt.title("Correlation Heatmap")
plt.show()
```



Which feature most affects survival?

Based on the visualizations and correlation analysis, **Sex** appears to have the strongest influence on passenger survival. Female passengers had a noticeably higher survival rate than male passengers. This aligns with the historical "women and children first" evacuation policy followed during the Titanic disaster.

Passenger class (Pclass) also shows a significant relationship with survival, indicating that first-class passengers generally had a higher chance of surviving than those in lower classes.

Conclusion

The Titanic dataset was successfully cleaned by handling missing values using appropriate techniques for numerical and categorical features. Visualizations provided valuable insights into the distribution of data, potential outliers, and relationships among variables.

The analysis indicates that passenger gender and class were among the most influential factors affecting survival. These findings provide a solid foundation for the predictive modeling tasks in the upcoming weeks.